fn ensure_open_unit_interval

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Proves soundness of `fn ensure_open_unit_interval` in `mod.rs`.

1 Hoare Triple

Precondition

None.

Pseudocode

```
1 def ensure_open_unit_interval(x: float):
2     if x <= 0.0 or x >= 1.0:  #
3         raise "failed to compute the repetition distribution parameters"
4     return ()
```

Postcondition

Theorem 1.1. For any input `x`, `ensure_open_unit_interval(x)` either returns `Err(e)`, or returns `Ok(())` and `x` lies in the open unit interval $(0, 1)$.

2 Proof

Assume an arbitrary input `x`.

Lemma 2.1. `ensure_open_unit_interval` returns `Err(e)` exactly when the guard on line 2 is true.

Proof. The guard is the only branch in the function. If it is true, the function executes the explicit `raise`. If it is false, the function reaches the final `return ()`. □

Lemma 2.2. If `ensure_open_unit_interval(x)` returns `Ok(())`, then `x` lies in $(0, 1)$.

Proof. If the function returns `Ok(())`, then by Lemma 2.1 the guard on line 2 is false. Therefore `x` is neither ≤ 0 nor ≥ 1, so `x` lies in $(0, 1)$. □

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. □